

COVID-19 IN CHILDREN

----- Active space -----

Guidelines for COVID-19 given by ministry of Health & Family Welfare (MOFW).

COVID appropriate behaviour :

3Ws & 2Vs.

- Watch your distance (>2 metres).
- Wear mask.
- Wash your hands frequently.
- Ventilation (open space ventilation ↓ transmission).
- Vaccination (for 15-18 years of age).

COVID-19 features in children

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Symptoms :

- mild/asymptomatic (m/c; incidental detection or h/o contact)
- Fever, cough, rhinorrhoea.
- Sore throat/ Throat irritation : **Omicron variant** (m/c).
- myalgia/headache /malaise.
- Diarrhoea.
- Loss of sense of smell > taste : **Delta variant** (m/c).

	Asymptomatic	mild (URT)	moderate	Severe
Respiratory rate	Normal	Normal	↑	↑
SPO ₂ in room air	21-94% khanirfan0392@gmail.com	21-94%	90-93%	< 90%
Danger signs/ symptoms	Absent	Absent	Absent	Present

Danger signs & symptoms :

- Grunting,
- Lethargy,
- Seizures, somnolence,
- Severe chest retractions.

Criteria for tachypnea based on age :

Age	Respiratory rate
< 2 months	> 60/min
2-12 months	> 50/min
1-5 years	> 40/min
> 5 years	> 30/min

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Management

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Investigations :

Severity of disease	Investigations
Asymptomatic	Not required
mild khanirfan0392@gmail.com	
moderate	CBC, ESR, Chest X-ray, Blood glucose.
Severe	• CBC, ESR, Chest X-ray, Blood glucose. • CRP/ferritin (markers of inflammation), • D-Dimer assays (for thrombosis), • LFT/RFT (for multi-organ dysfunction).

CT chest :

- **Not indicated** in paediatric covid.
- Except if **terminally ill**.

Treatment :

	Asymptomatic	mild	moderate	Severe
Site of care	Home isolation	• Home isolation ± Teleconsultation (sos) • Covid care centre	• Ward of a COVID 19 hospital • DCHC (Dedicated COVID Health Centre).	• ICU • High Dependency unit (HDU) of COVID-19 hospital.
Rx	No specific Rx	Symptomatic Rx : • Paracetamol for fever. • Warm saline gargles for sore throat.	• O₂ inhalation (target SpO₂ : 94-96%). • Fluid & electrolyte balance correction. • Steroids : in progressive disease course. • Antimicrobials : If needed.	• O₂ support (SpO₂ : 94-96%). • Fluid & electrolytes balance correction. • Corticosteroids : - most useful drug - Indicated in all severe cases.

Steroid administration :

- Right time : **Avoided in first 3 - 5 days**, after onset of symptoms (prolongs the viral shedding.)
- Right dose :
 1. **Dexamethasone** : 0.15 - 6mg/kg once daily.
 2. **methyl prednisolone** : 0.75 - 30mg/kg once daily.
- Right Duration : 5 - 7 days → tapered over 10 -14 days.

Anticoagulants :

- Indication : thrombosis confirmed by imaging/ $\uparrow$ D-dimer levels.
- Low molecular weight heparin /LMWH (Enoxaparin) : 1 mg/kg BD.

Antimicrobials : Indications

- Clinical evidence/ Suspicion of bacterial infection.
- Septic shock.

Antivirals (Remdesivir, Favipiravir)/monoclonal antibodies :

Not recommended (efficacy unknown in children.)

Multisystem inflammatory syndrome in children (Mis-C)

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Time of presentation : 2-4 weeks after acute COVID.

Risk factors :

- Age < 1 year.
- Co-morbidities : Asthma, Chronic lung disease.
- Obesity.
- Children with immunodeficiency.

Diagnostic criteria (WHO) for mis-C :

1. Children and adolescents with fever ($>38^{\circ}\text{C}$) for ≥ 3 days. AND
2. Any 2 of the following should be present : Evidence of
 - mucocutaneous inflammation : Rash/ Bilateral non purulent conjunctivitis.
 - Hypotension or shock.
 - myocardial dysfunction, myocarditis, valvulitis, Coronary aneurysms, $\uparrow$ troponin.
 - Coagulopathy (elevated D-dimers).
 - Severe GI involvement (uncontrolled abdominal pain /diarrhoea).
3. And additional findings :
 - a. $\uparrow$ inflammatory markers :
 - CRP > 5 mg/dl.
 - ESR > 40 mm/hr.
 - Procalcitonin.
 - b. Recent COVID-19 (RTPCR +ve)/ Contact with confirmed COVID-19 case.
 - c. No evidence of serious bacterial infection/ Alternative diagnosis.

management of mis-C :

- Hospitalization in ICU.
- methyl prednisolone (2mg/kg/day) : tapered over 2-3 weeks.
- IV Immunoglobulin (IVIg) :
 - Dose : 2g/kg as slow IV infusion.
 - maximum dosage : 100g.

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